

ABHISHEK LOHAR

SOFTWARE DEVELOPMENT • AI/ML SYSTEMS • FULL-STACK • BACKEND ARCHITECTURE

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PROFESSIONAL SUMMARY

Innovative and results-driven Computer Science student at VIT-AP (CGPA: 8.33) with comprehensive expertise spanning **production software engineering, distributed backend systems, AI/ML pipelines, and full-stack development**. Author of a published open-source PyPI package with 160 unit/integration tests and automated CI/CD pipelines. Demonstrated ability to engineer low-latency architectures with Redis, Kafka, and FastAPI, alongside edge computer vision with OpenCV and Raspberry Pi. Adept in algorithmic problem-solving, clean code design, and cross-functional team delivery.

CORE TECHNICAL COMPETENCIES

Programming Languages:	Java, Python, C++, SQL (PostgreSQL), JavaScript, HTML5/CSS3
Backend & Distributed Systems:	FastAPI, RESTful APIs, Microservices, Redis (Caching), Apache Kafka, API Gateway
AI, ML & Vector Search:	Ollama, LangChain, ChromaDB, RAG Pipelines, Semantic Search, LLMs, OpenCV, Embeddings
DevOps, Testing & Tools:	Git, GitHub Actions CI/CD, PyPI Package Deployment, pytest (160+ tests), Linux/Bash, Raspberry Pi
Core CS Fundamentals:	Data Structures & Algorithms, Object-Oriented Design (OOD), DBMS, Operating Systems, Computer Networks

EDUCATION

Vellore Institute of Technology-AP (VIT-AP) — Amaravati, India

2023 – 2027

Bachelor of Technology in Computer Science and Engineering — **CGPA: 8.33 / 10.0**

- **Relevant Coursework:** Data Structures & Algorithms, Database Systems, Object-Oriented Programming, AI Foundations.

Vidya Bharati Chinmaya Vidyalaya (CBSE) — Jamshedpur, India

2022

Higher Secondary Education (Class XII) — **Aggregate: 79.8%**

FEATURED ENGINEERING PROJECTS

local-persona-memory — Production Open-Source Python Package (PyPI) | *Python, LangChain, ChromaDB, Ollama, RAG, pytest, GitHub Actions*

- Architected and published a production-grade Python package on **PyPI** (pip install local-persona-memory), giving local LLMs autonomous long-term persistent memory with 100% data privacy and zero cloud dependence.
- Designed a resilient RAG pipeline incorporating ChromaDB vector embeddings, chunked PDF/document ingestion, sub-second semantic search, and dynamic conversation recall hooks.
- Implemented **160 passing automated unit/integration tests** with pytest and established end-to-end automated CI/CD workflows with GitHub Actions for release quality assurance.

Distributed URL Shortener & Analytics Engine | *FastAPI, PostgreSQL, Redis, Apache Kafka, Microservices Architecture*

- Developed a decoupled, scalable microservices backend featuring isolated Shorten, Redirect, and Analytics services coordinated behind a unified API Gateway.
- Configured **Redis in-memory caching** for sub-5ms URL redirects, paired with **PostgreSQL** for ACID-compliant persistence and transactional durability.
- Integrated **Apache Kafka** event streaming to asynchronously decouple incoming click metrics from the critical redirect path, guaranteeing zero latency penalty for end-users.

Smart Attendance Face Recognition System | *Raspberry Pi, OpenCV, Python, Computer Vision, SQLite/MySQL*

- Built an end-to-end embedded biometric attendance device integrating Raspberry Pi hardware cameras with real-time OpenCV detection.
- Trained facial landmark classification models to automate contactless attendance logging with **95%+ recognition accuracy**.
- Optimized frame-processing throughput for resource-constrained edge hardware, preventing frame drops during concurrent verification.

CERTIFICATIONS, HONORS & AWARDS

- **Oracle Cloud Infrastructure (OCI) Certified AI Foundations Associate:** Validated enterprise AI and cloud paradigms.
- **Coursera Full Stack Web Development Specialization:** Hands-on modern responsive web systems and REST architecture.
- **Infosys Springboard — Introduction to Data Science:** Advanced data wrangling, analytics, and statistical modeling.
- **High-Performance Coding (HPC) Certification (IAMNEO):** Algorithmic problem solving and runtime execution optimization.
- **Hashgraph Developer Course:** Distributed ledger technology, consensus algorithms, and decentralized applications.
- **Finalist, Engineering Clinics 2025:** Recognized among top engineering innovators for practical hardware-software solutions.
- **Active Competitor, Hack2Skill GDG Hackathons:** Accelerated collaborative prototyping under competitive time constraints.

LANGUAGES & INTERESTS

Languages: English (Professional Working Proficiency), Hindi (Native) | **Interests:** Open Source Development, LLM Autonomous Agents, Edge AI, Distributed Systems